

## Assignment 4

### Neuromembrane Lab 4

The purpose of this assignment is to help you understand how membrane permeability and ion concentrations contribute to the resting membrane potential. We will go through multiple simulations. **Be sure to answer all the questions throughout this assignment.** The questions are marked with this symbol 

#### Goals:

1. To examine the refractory period following the action potential and its consequences for membrane excitability
2. To understand how the resting potential depends on the relative permeability (conductances) of  $\text{Na}^+$  and  $\text{K}^+$ .
3. To understand how signals can be generated by changing the ratio of the conductances to Na (gNa) and K (gK).
4. To experiment with neurotoxins and observe their effects on the action potential.

### *Action Potentials using Neuromembrane Simulator*

<https://neuromembrane.biology.ualberta.ca/>

#### How to use the program:

1. Select “Log in as a guest” button.
2. You’ll now be inside of the action potential simulation.
  - a. **Tip:** Double-check that this is the simulation you’re in by looking at the top of the panel to the left. It should say “Action Potential Simulation.”
3. **Main page sections:**
  - **Left Section:** Allows you to adjust the parameters necessary for the simulation and is composed of several boxes that allow you to control the leak channels,  $\text{Na}^+/\text{K}^+$  pump, initial membrane voltage, ion concentration settings, etc.
  - **Middle Section:** Shows a plain lipid bilayer upon which you can build your excitable membrane. Clicking the “+” sign at the top right of the box adds that particular item to the membrane in the right half of the window and allows you to adjust more detailed parameters for that item. These parameters can be adjusted by either clicking on the number, directly typing it in, or by dragging the slider left and right. The small counterclockwise arrow in the bottom right hand corner of each box allows you to revert back to default values.
  - **Right Section:** Shows the membrane simulation. The  $\text{Na}^+$  and  $\text{K}^+$  ions can be seen in the top half of the window, which represents the intracellular side. Adjusting the concentrations of the ions results in a greater (or lower) number of ions in the appropriate field of view.

## Experiment 1 – Refractory Period

**Background:** It is well known that there is a period of reduced excitability following an action potential called the "refractory period." During this time, lingering changes in the ionic conductance following one impulse affect the threshold for firing a next impulse. The refractory period limits the frequency of impulses in a train of impulses.

*Steps to complete Experiment 1:*

1. Let's add ion channels and the sodium-potassium pump! Within the control panel on the left-hand side, you'll see several components that you can add to your cell membrane. These components include leak channels, Na<sup>+</sup> channel, K<sup>+</sup> channel, and the Na<sup>+</sup>/K<sup>+</sup> pump. **Add all these components by clicking the + button.** See screenshot below.
2. Now that these elements are reflected on your cell membrane, let's adjust the "external positive current" that you will be applying to the membrane through an electrode. Change the external current panel so that you inject **TWO separate currents at an intensity of 25  $\mu$ A**. It should now look like this:

| Current ( $\mu$ A) | Time (ms) |
|--------------------|-----------|
| 0                  | 10        |
| <b>25</b>          | 2         |
| 0                  | 10        |
| <b>25</b>          | 2         |
| 0                  | 16        |

3. We will now decrease the "inter-current interval" (time between the two stimuli) from 10 ms to 0.25 ms. It should now look like this:

| Current ( $\mu$ A) | Time (ms)   |
|--------------------|-------------|
| 0                  | 10          |
| <b>25</b>          | 2           |
| 0                  | <b>0.25</b> |
| <b>25</b>          | 2           |
| 0                  | 16          |



4. Now run the simulation! Describe what happened regarding the number of currents and number of action potentials. Was each stimulating current followed by an action potential?

**An action potential was generated after only the first current was administered, and not for the second current which was administered. The second current was administered at the same time the action potential was undergoing "hyperpolarization" and had not yet returned to**

homeostasis/baseline (so another action potential would not have even been possible as the glial cell was still carrying out the action potential created by the first current administered.

5. How could you determine whether under these conditions the neuron is in absolute or relative refractory period? Go back to settings and increase the intensity of the second stimulus in the “External Current” section to test your hypothesis (it doesn’t matter what values you select, just pick a few and see if the increased stimulus intensity results in an AP or not).

| Current ( $\mu\text{A}$ ) | Time (ms) |
|---------------------------|-----------|
| 0                         | 10        |
| 25                        | 2         |
| 0                         | 0.25      |
| 75                        | 2         |
| 0                         | 16        |



- a. If you increase the intensity of the second stimulus, do you get a second action potential? Yes or no? Does this represent the absolute or relative refractory period? Explain why.

No, increasing the intensity of the second stimulus still results in only a single action potential. This is an example representative of the “absolute refractory period” concept because regardless of the second stimulus’s intensity, it is too soon (.25ms) in sequence to the first current and therefore another action potential can never be generated.

- b. What prevents another action potential from occurring in the repolarization phase of the first action potential?

The voltage-gated sodium channels are in an inactive state and not yet able to close and then open again, which would allow the second influx of sodium ions required for another action potential to occur.

6. Go back to the “External Current” settings and increase the “inter-current interval” (time between the two stimuli) from 0.25 ms to 3 ms. It should now look like this the values according to the table below:

| Current ( $\mu\text{A}$ ) | Time (ms) |
|---------------------------|-----------|
| 0                         | 10        |
| 25                        | 2         |
| 0                         | 3         |

|    |    |
|----|----|
| 25 | 2  |
| 0  | 16 |



7. You have now increased the inter-current interval from 0.25 ms to 3 ms. Run the simulation to see what happens when the second current is applied. Explain what you see.

The graph for the 3ms interval is very similar to that of graph for the .25ms interval. However, one key noticeable difference here is that in this trial, the voltage slightly jumps above the resting potential of -70.4mV up to -66.9 mV before coming back down to resting potential again (which indicates the second action potential wanted to occur this time, and was more likely to occur given the larger time interval between currents, but was still unable to actually occur).

8. Finally, determine again whether under these conditions the neuron is in absolute or relative refractory period. To do so, go back to settings and increase the intensity of the second stimulus in the “External Current” section to test your hypothesis (it doesn’t matter what values you select, just pick a few and see if the increased stimulus intensity results in an AP or not).

| Current ( $\mu$ A) | Time (ms) |
|--------------------|-----------|
| 0                  | 10        |
| 25                 | 2         |
| 0                  | 3         |
| 75                 | 2         |
| 0                  | 16        |



- a. If you increase the intensity of the second stimulus, do you get a second action potential? Yes or no? Does this represent the absolute or relative refractory period? Explain why.

Yes, increasing the stimulus intensity three-fold results in an action potential for this 3ms time interval. This is representative of the concept for the “relative refractory period” because the 3ms interval is enough time (compared to .25ms) to allow the sodium voltage-gated ion channels to close (and therefore can be opened again). In addition to the time interval, the stimulus intensity meets the minimum voltage required to open these channels and allow the influx of sodium ions.

- b. Why is a greater stimulus required to obtain a second action potential? Include commentary about  $\text{Na}^+$  and  $\text{K}^+$  channels in your answer.

A greater stimulus is required to obtain a second action potential because the glial cell is still in the hyperpolarization phase (ion channels for  $\text{Na}^+$  still not all fully closed and

*K<sup>+</sup> still open). This means that the neuronal membrane is increasingly negative (in the -80 to -90mV range compared to the usual resting point of -70mV, which makes it harder to fire again) due to the potassium efflux and as a consequence a larger stimulus intensity is required to bring the voltage to its firing threshold (-55mV).*

- c. Was the amplitude of the action potentials the same? Why?

*No, the amplitude of the first and second action potential are not equivalent. The amplitude of the first AP was greater than the second because not all the Na<sup>+</sup> channels had closed yet by the time the second stimulus signal arrived, so they were not all able to open again and thus the maximum voltage potential could not be reached again like seen with the first AP (we need all Na<sup>+</sup> gates fully functional for the maximum amplitude of an AP to occur)*

## ***Tutorial 2 – The Ca Action Potential***

### **Background:**

#### ***Voltage-gated Ca channels can support action potentials.***

Since the 1950's we have known that certain cell types can generate Ca-dependent action potentials. Voltage-gated Ca channels, rather than Na channels, underlie the regenerative rising phase of some action potentials and prolong the falling phase of others. The driving force on Ca is enormous because of its extraordinarily low concentration inside the cell. Consequently, opening Ca channels can result in a huge inward Ca current.

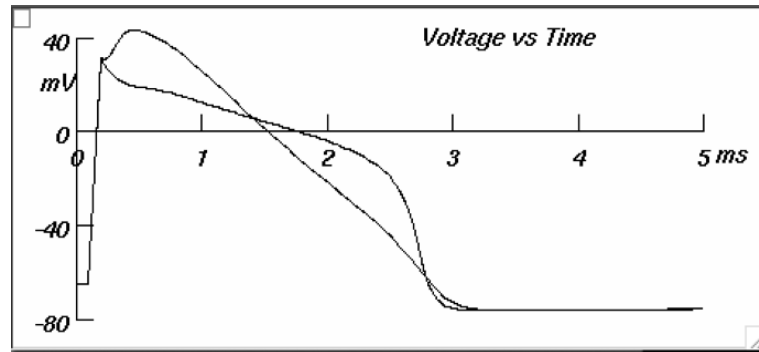
#### ***Ca channels comprise many subtypes.***

Many subtypes of voltage-gated Ca channels have now been identified, revealed by molecular biology and patch clamping. A lucid summary of the identity and properties of these channels and their families as of 2001 can be found in Hille (2001). In this tutorial you will use a commonly-found channel, the high-voltage-activated L-type Ca channel from the CA3 pyramidal neuron of the hippocampus (Migliore et al., 1995). This channel has been imported into NIA from the ModelDB (modeling database) of the NEURON website.

#### ***Cardiac action potentials depend on the properties of Ca channels.***

Ca entry during action potentials, through a variety of Ca channel subtypes, is used to trigger or regulate various cell functions. Examples include the secretion of hormones and neurotransmitters, muscle contraction, and pacemaking. In this tutorial you will experiment with prolonged action potentials to simulate those in cardiac muscle. Although the action potentials in cardiac pacemakers and in ventricular myocytes are extraordinarily complex, involving many different ion channels and pumps, and cannot be simulated here with precision, it is possible to gain an understanding of how the presence of Ca channels in these cells gives rise to the long and unusual cardiac action potentials.

Compare  $\text{Na}^+$  to  $\text{Ca}^{2+}$  action potentials. See image below from the lecture slides, which depicts both a  $\text{Na}^+$  and  $\text{Ca}^{2+}$  action potential, to help you visualize the difference ( $\text{Na}^+$  is taller,  $\text{Ca}^{2+}$  is wider).



1. According to the description above, as well as the lecture slides, what type of cells/cell functions are  $\text{Ca}^{2+}$  action potentials most commonly involved in?

These types of action potentials are commonly found in:

1. Endocrine cells for secretion of hormones
2. Neuronal cells for neurotransmitter release
3. Visual transduction in rod photoreceptors
4. Muscle contraction (skeletal, smooth, contractile myocardial cells)
5. Cardiac pacemakers (autorhythmic cells)



2. According to the description above, as well as the lecture slides, summarize the major differences between  $\text{Na}^+$  and  $\text{Ca}^{2+}$  action potentials.
  - a. Which one is quicker to reach threshold?

$\text{Na}^+$  action potentials are faster (reach their threshold quicker) than  $\text{Ca}^{2+}$  action potentials

- b. Which one takes longer to return to resting potential? Why?

$\text{Ca}^{2+}$  action potentials take longer to return to resting potential because the  $\text{Ca}^{2+}$  ion gates are slower to inactivate (close). Also, we can see from the graph above that the influx of ions is sustained/prolonged at a slower and constant rate (giving us the characteristics of a “longer” action potential)

- c. Why is the peak amplitude of the calcium action potential shorter than that of the sodium action potential?

This is due to the key difference in the mechanics of how  $\text{Na}^+$  and  $\text{Ca}^{2+}$  action potentials occur. For  $\text{Na}^+$  action potentials, the influx of  $\text{Na}^+$  ions and the efflux of  $\text{K}^+$  ions occur in sequence but at separate times (and the influx of  $\text{Na}^+$  ions is very rapid, resulting in the sudden spike seen on the graph). For  $\text{Ca}^{2+}$  action potentials, the influx of  $\text{Ca}^{2+}$  ions and

efflux of  $K^+$  ions occur simultaneously and slowly, so without the rapid influx of ions (which DOES occur in  $Na^+$  action potentials) we will not see a sudden spike on the graph (and indeed we do not)

## ***Tutorial 3 – The Neuromuscular Junction***

### **Background:**

***This tutorial simulates the responses of a muscle fiber to acetylcholine (ACh) released from the motoneuron's presynaptic terminal.***

The simulations model the postsynaptic (or "end plate") membrane of the well-known neuromuscular junction (NMJ) of the frog. At the NMJ, ACh binds to its postsynaptic (nicotinic) receptors, which are also ion channels. The channels open, causing a postsynaptic "end plate current" (EPC) to flow that leads to a postsynaptic, depolarizing "end plate potential" (EPP) in the muscle fiber.

***ACh-gated receptors are permeable to both Na and K ions.***

Activation of ACh receptors causes a simultaneous and equal increase in the postsynaptic conductance to both  $\text{Na}^+$  and  $\text{K}^+$  ions. In this tutorial you will see the relationships between the ACh-gated conductance, the resulting EPC, and the EPP.

***Studying the NMJ reveals general principles of excitatory synapses.***

In later tutorials you will experiment with excitatory postsynaptic potentials (EPSPs) in neurons, such as: the interaction of EPSPs in a patch, the spread of an EPSP along a dendrite to the soma and axon, and the temporal integration of more than one EPSP.

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1. What is the neuromuscular junction? Which neurotransmitter is most important and what kind of receptor does it bind to?

***The NMJ is the point at which the terminal end of a motor nerve synapses onto a muscle (this is where a cognitive “goal-plan” turns into an actual behavior/motion. The important neurotransmitter here is Acetylcholine as it plays the essential role of binding to postsynaptic nicotinic receptors (ion channels) which open and allow the influx of  $\text{Na}^+$  and efflux of  $\text{K}^+$  necessary for a EPP to occur in the muscle fiber***

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2. What direction is the flow of  $\text{Na}^+$  and  $\text{K}^+$  through the receptors in question 1? Which ion conductance is greater?

***Similar to an AP, the sodium ions flow into the cell and potassium leaves the cell through the receptors in question #1. Additionally, the influx of sodium ions is greater than potassium's efflux rate, so the cell depolarizes as a result (and an EPP is more likely to occur).***

## Experiment 4 – Interactions of Synaptic Potentials

### Background:

#### *How do synaptic potentials interact?*

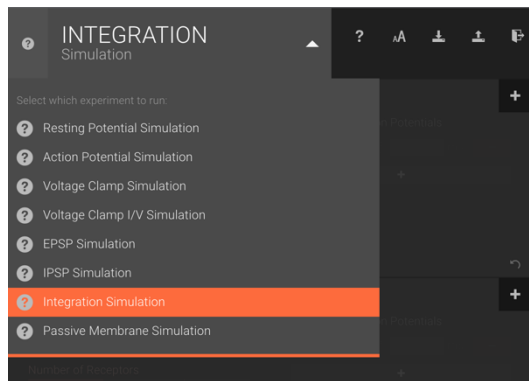
Can an excitatory postsynaptic potential (EPSP) or an inhibitory postsynaptic potential (IPSP) influence the subsequent excitability of a neuron? In this experiment you will investigate why these synaptic potentials can have an impact on whether or not a subsequent EPSP causes a neuron to fire.

#### *This experiment focuses on "temporal summation," the interaction of EPSPs and IPSPs in time.*

These experiments explore the interaction of synaptic potentials that essentially interact at the same point. Two EPSPs might come from two spatially close presynaptic inputs or from two sequential action potentials in the same neuron. An EPSP and an IPSP would naturally come from an excitatory and an inhibitory input located close to one another on the cell.

#### *Steps to complete Experiment 4:*

1. Log out of the neuromembrane simulation and log back in as a “guest”. This will reset all your settings.
2. You will want to go up to the top drop-down menu and change the experiment type to “Integration Simulation”. See screenshot below.



3. This simulation allows you to add up to a maximum of 3 synapses on to a cell body, to examine how excitatory and inhibitory synapses sum, or integrate. Clicking the "plus" sign at the top right of each box adds a synapse to the cell. You have the option of making the synapse excitatory (EPSP) or inhibitory (IPSP) under the "Type of Synapse" setting. Inhibitory synapses are assumed to contain GABA receptors, as in the IPSP Simulation. You can change the single channel conductance and the number of receptors at each synapse. Additionally, you can determine when that synapse becomes active by adjusting the time at which an action potential is generated presynaptically. This allows you to determine how the timing of excitatory and inhibitory events integrate to result in an action potential, or not.

4. To begin, add “Synapse 1” to the cell body by clicking the “plus sign” at the top right of the box. Run the simulation. Keep the timing of the action potential at 10 ms.



a. Did you get an action potential? YES or NO?

NO



b. What was the max voltage? -65.40 mV

5. Now, add “Synapse 2” to the cell body by clicking the “plus sign” at the top right of the second box. You should now have **two** synapses. Keep the timing of the action potentials from both synapses at 10 ms. This means that both synapses will send an excitatory current at the same time. Run the simulation.



a. Did you get an action potential? YES or NO?

YES



b. What was the max voltage? 38.60 mV

6. Now, add “Synapse 3” to the cell body by clicking the “plus sign” at the top right of the second box. You should now have **three** synapses. Keep the timing of the action potentials from all synapses at 10 ms. This means that all synapses will send an excitatory current at the same time. Run the simulation.



a. Did you get an action potential? YES or NO?

YES



b. What was the max voltage? 41.90 mV

7. What does this experiment reveal about the interaction of EPSPs? How does this demonstrate “summation”? Explain what is happening.

When there is a single synapse on a neuron, that one input/signal itself is not sufficient to surpass the absolute threshold. When there is a second synapse onto the same neuron, there is now an additional opportunity for energy input which will be sufficient enough if the two EPSP's summate temporally or spatially. Finally, with a third synapse/EPSP added, the graph indicates that there is still an increase in the maximum voltage, although it is greatly reduced due to the finite number of ions and channels available at that time (cell can only do so much in a given period of time). Also demonstrated was the importance of timing/arrival of these signals

8. Now, go back to the settings page and change one of the synapses to “IPSP”. It doesn’t matter which one, you just need to have two EPSP synapses (excitatory) and 1 IPSP (inhibitory). Keep the timing of the action potentials from all synapses at 10 ms. This means that all synapses will send either an excitatory or inhibitory current at the same time. Run the simulation.



- a. Did you get an action potential? YES or NO?

NO



- b. If yes, what was the max voltage? -62.60 mV



9. What does this experiment reveal about the interaction of EPSPs AND IPSPs? Explain what is happening.

*This experiment reveals that 2 EPSP’s and IPSP’s will influence each other if they summate spatially. More specifically, the single IPSP cancels out (or will at least inhibit) one of the EPSP’s leaving only one single EPSP which we saw earlier was not sufficient itself to generate an AP. All of this is dependent though on simultaneous signaling from all 3 of the synapses*



10. What is the difference between temporal and spatial summation (see last slide of lecture). What type of summation did we witness in this demonstration?

*In this demonstration we witnessed spatial summation. The difference between the two is that temporal summation involves 2 “excitatory” stimuli whereas spatial summation can be the sum of both an excitatory and inhibitory signal. Additionally, temporal summation must only occur in “close” proximal time whereas in spatial summation the signals must occur simultaneously*